

HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 13

March 10, 2026



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Quiz 6

The quizzes are designed to take no more than 30 minutes, but one hour will be allowed.

Do not click into the quiz until you are ready to take it.



CS-2050 / homework-3

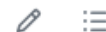
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question-7	public release.	4 days ago
README.md	public release.	4 days ago

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Homework 3

Welcome to Homework 3. This assignment is due by 11:59pm on **Thursday March 12**. Begin with the setup instructions in the next section, and don't forget the submission instructions at the bottom.

About

No description, website, or topics provided.

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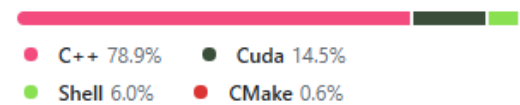
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Contributors 2

- wcw398 Chuck Witt
- laz380 Laura Zichi

Languages



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wcw398

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4be70a2 · 17 hours ago

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README

Questions and Answers

Please create issues in this repository to ask questions about the course.

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About

About

Ask your CS 2050 questions here.

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 README.md draft release. 2 minutes ago

CS 2050 Final Project

Warning

This is a DRAFT of the final project specification, as of *March 5, 2026*. Please review and ask any clarification questions over the next two weeks. Based on feedback, we will refine and release the final specification at the end of the Spring Recess.

Overview

The final project requires you to apply the high-performance computing techniques learned in class to a computational task of your choice. The project will be completed **individually**. You will begin with a serial version of your algorithm, and then develop parallel versions using OpenMP, MPI, and CUDA. You will analyze its performance in scaling as well as with profiling tools. Finally, you will explore an additional topic beyond the core course material.

High Performance Computing for Science and Engineering

CS 2050 Challenges

Reminder: Your pseudonym can be found with ``challenge_pseudonym`` on the course cluster.

Last Update: Thursday, Mar 05, 1:51:27 PM

Leaderboards

challenge_0

0 ranked submissions

No results yet.

RULES FOR TODAY

- Group work is encouraged
- No AI tools allowed for the first 10 minutes after each new question release

SETUP

- Request a full node for interactive use

```
srun --nodes=1 --exclusive --time=01:00:00 --pty bash
```

- Load the CS-2050 environment

```
. ~/161588/spack/share/spack/setup-env.sh  
spack env activate -p CS-2050
```


CHALLENGE 0

```
double compute(pybind11::array_t<double> &X_in)
{
    const int N = X_in.size();    // size of array
    const double *X = X_in.data(); // pointer to first element

    // Compute the result
    double F = 0.0;
    for (int i = 0; i < N; ++i)
    {
        for (int j = 0; j < N; ++j)
        {
            for (int k = 0; k < N; ++k)
            {
                if (i == j or j == k or k == i)
                    continue;
                F += X[i] * X[j] * X[k];
            }
        }
    }
    return F;
}
```

CHALLENGE 1

```
double compute(pybind11::array_t<double> &X_in)
{
    const int N = X_in.size();    // size of array
    const double *X = X_in.data(); // pointer to first element

    // Compute the result
    double F = 0.0;
    for (int i = 0; i < N; ++i)
    {
        for (int j = i; j < N; ++j)
        {
            for (int k = j; k < N; ++k)
            {
                for (int l = k; l < N; ++l)
                {
                    F += X[i] * X[j] * X[k] * X[l];
                }
            }
        }
    }
    return F;
}
```

CHALLENGE 2

```
double compute(const int num_i,
               std::vector<int> num_j,
               pybind11::array_t<double> &X_in)
{
    double *X = X_in.mutable_data();

    double result = 0.0;
    for (int i = 0; i < num_i; ++i)
    {
        for (int j = 0; j < num_j[i]; ++j)
        {
            for (int k = 0; k < j; ++k)
            {
                result += j * k * X[j] * X[k];
            }
        }
        X += num_j[i];
    }
    return result;
}
```

CHALLENGE 3

```
double compute(pybind11::array_t<double> &X_in,
               pybind11::array_t<int> &index_1_in,
               pybind11::array_t<int> &index_2_in,
               pybind11::array_t<double> &C_in)
{
    const int M = X_in.request().shape[0]; // num_rows
    const int N = X_in.request().shape[1]; // num_columns
    const double *X = X_in.data();
    const int num_multiplications = index_1_in.size();
    const int *index_1 = index_1_in.data();
    const int *index_2 = index_2_in.data();
    const double *C = C_in.data();

    double result = 0.0;

    for (int i = 0; i < M; ++i)
    {

        double Y[N + num_multiplications];

        for (int j = 0; j < N; ++j)
        {
            Y[j] = X[i * N + j];
        }
    }
}
```

CHALLENGE 4

```
void strided_math(const int stride,
                  const pybind11::array_t<float>& X_in,
                  pybind11::array_t<float>& Y_in)
{
    const float* X = X_in.data();
    float* Y = Y_in.mutable_data();

    for (int i=0; i<Y_in.size(); ++i) {
        float x = X[i*stride];
        Y[i] = std::exp(std::sin(std::sqrt(x) + std::cos(x*x)))
               * std::cos(std::exp(-x) + std::exp(-std::sqrt(x+1.0)))
               * (std::sin(std::exp(x)) + std::cos(std::exp(-x)))
               * (std::sin(std::sqrt(x+2.0)) + std::cos(std::sqrt(x+3.0)));
    }
}
```

CHALLENGE 5

```
double compute(const pybind11::array_t<double> &data_in)
{
    const double *data = data_in.data();
    const int n = data_in.size();

    double sum = 0.0;
    double sum_of_squares = 0.0;
    double max = 0.0;
    #pragma omp parallel for
        for (int i = 0; i < n; ++i)
        {
            #pragma omp critical
            {
                sum += data[i];
                sum_of_squares += data[i] * data[i];
                if (data[i] > max)
                {
                    max = data[i];
                }
            }
        }

    double total_log = 0.0;
    #pragma omp parallel for
```